Chapter-01
Introduction

In this chapter, we will discuss about the origin of statistics with its scopes, functions, and limitations.

1.1. Origin and Historical Development of Statistics

The development of statistics can be traced to ancient civilizations, where numerical records were maintained for population, taxation, agriculture, trade, and military administration. Over time, these practices evolved into systematic methods for collecting and interpreting quantitative information. During the seventeenth century, the work of John Graunt on mortality records represented an important step toward the systematic analysis of demographic data. During the same period, Blaise Pascal and Pierre de Fermat laid important foundations for the mathematical theory of probability, establishing a close relationship between probability and statistics.

During the eighteenth and nineteenth centuries, statistics gradually developed as an academic and mathematical discipline. Gottfried Achenwall contributed to the systematic use of the term *Statistik*, while the work of Carl Friedrich Gauss on least squares and the normal distribution strengthened the mathematical foundations of statistical analysis. Adolphe Quetelet applied statistical methods to social and biological phenomena, whereas Francis Galton contributed significantly to the development of correlation and regression.

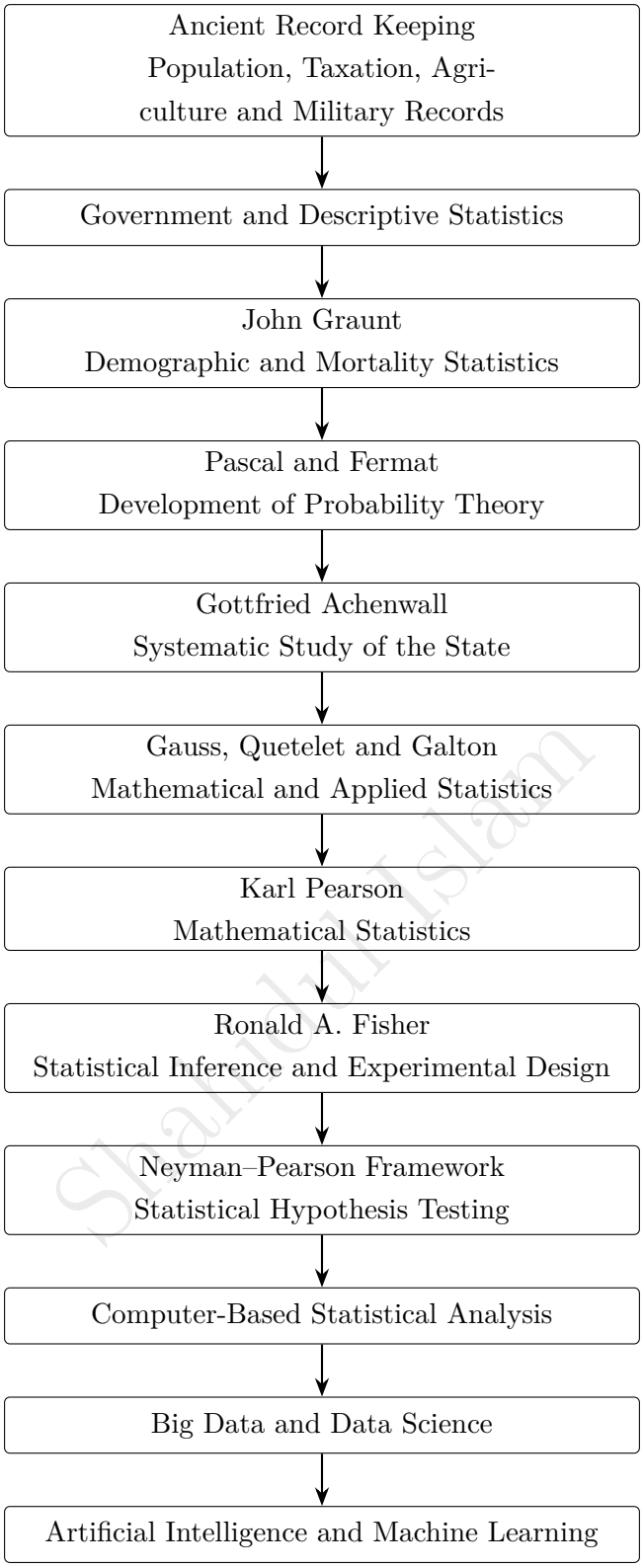
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In the late nineteenth and early twentieth centuries, Karl Pearson further established mathematical statistics through his contributions to correlation, statistical distributions, curve fitting, and the chi-square test. During the twentieth century, Ronald A. Fisher made fundamental contributions to statistical inference, experimental design, analysis of variance, and maximum likelihood estimation. Subsequently, the Neyman–Pearson framework provided a rigorous foundation for statistical hypothesis testing.

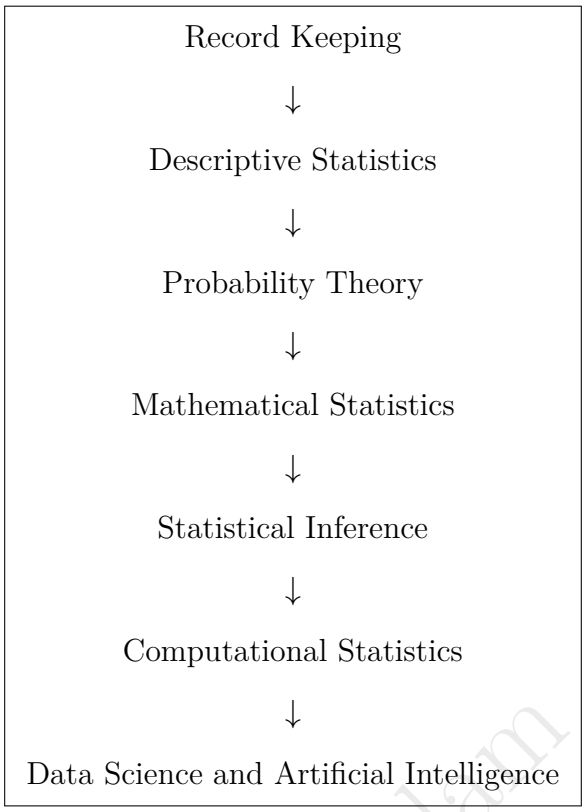
The emergence of electronic computers transformed statistical practice by enabling the analysis of large and complex datasets. In the twenty-first century, statistics has become an essential foundation of **Data Science, Artificial Intelligence, Machine Learning**, and other computational disciplines. Thus, the historical development of statistics represents a continuous transition from **record keeping and descriptive information** to **mathematical modeling, statistical inference, computational analysis, and data-driven decision-making**.

Historical Evolution of Statistics

The major stages in the historical development of statistics can be summarized as follows:



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1.2. Meaning of Statistics

The term **Statistics** is used in two different senses: the **plural sense** and the **singular sense**. In the plural sense, statistics refers to numerical facts or data, whereas in the singular sense, it refers to the scientific methods used for collecting, organizing, presenting, analyzing, and interpreting data.

Statistics in the Plural Sense

In the plural sense, statistics refers to a collection of **numerical facts or observations** obtained for a particular purpose. Such numerical information may describe population, economic, social, scientific, educational, or technological phenomena.

For example:

- The population of a country is 170 million.
- The average marks of a class are 72.
- A computer network processes 10,000 packets per second.
- The failure rate of a software system is 2%.

In these examples, the numerical values themselves are regarded as **statistics in the plural sense**.

1.2.0.1 Statistics in the Singular Sense

In the singular sense, statistics is regarded as a **scientific discipline** concerned with the systematic study of data. It involves a sequence of activities through which raw data are converted into meaningful information.

The major activities of statistics are:

1. Collection of data;
2. Organization of data;
3. Presentation of data;
4. Analysis of data; and
5. Interpretation of data.

Thus, statistics provides systematic methods for obtaining useful information from data and making decisions in the presence of uncertainty.

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1.3. Definition of Statistics

A general definition of statistics may be stated as follows:

> **Statistics is the science of collecting, organizing, presenting, analyzing, and interpreting data for the purpose of drawing meaningful conclusions and making informed decisions.**

The statistical process can therefore be represented as

Data Collection → Organization → Presentation → Analysis → Interpretation → Decision Making

For Computer Science students, statistics provides the mathematical and analytical foundation for extracting useful information from data. It is widely used in areas such as:

- Artificial Intelligence and Machine Learning;
- Data Science and Data Mining;
- Computer Networks;
- Cybersecurity;
- Computer Vision;
- Natural Language Processing;
- Software Engineering; and

- Performance and Reliability Analysis.

For example, suppose a web server records the response time of 10,000 requests. The raw response times constitute **data**. Statistical methods can then be used to calculate the average response time, measure variability, identify unusual observations, and make predictions about future system performance. Therefore, for a Computer Science student,

$$\boxed{\text{Statistics} = \text{Data} + \text{Analysis} + \text{Interpretation} + \text{Decision Making}}$$

1.4. Types of Statistics

Statistics can be broadly classified into two main types:

1. **Descriptive Statistics**

   Descriptive statistics deals with the collection, organization, summarization, and presentation of data. It describes the main features of a dataset using tables, graphs, averages, percentages, and measures of dispersion.

   Examples: Mean, median, mode, range, variance, standard deviation, tables, charts, and graphs.

2. **Inferential Statistics**

   Inferential statistics deals with drawing conclusions or making predictions about a population based on information obtained from a sample. It uses probability and statistical methods to estimate population characteristics and test hypotheses.

   Examples: Estimation, confidence intervals, hypothesis testing, correlation, regression, and analysis of variance (ANOVA).

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$$\boxed{\text{Statistics} = \begin{cases} \text{Descriptive Statistics}, \\ \text{Inferential Statistics}. \end{cases}}$$

1.5. Importance of Statistics in Computer Science and Engineering

Statistics plays a fundamental role in Computer Science and Engineering because modern computing systems generate, process, and analyze large amounts of data. Statistical methods provide systematic techniques for understanding data, quantifying uncertainty, identifying patterns, evaluating performance, and making evidence-based decisions. The importance of statistics in Computer Science and Engineering can be understood through the following major areas.

1.5.1 Data Analysis and Data Science

Statistics provides the fundamental tools for collecting, organizing, summarizing, analyzing, and interpreting data. Measures such as the mean, median, variance, standard deviation, and correlation are widely used to understand the characteristics of datasets. In Data Science, statistical methods are used to transform raw data into meaningful information and support data-driven decision-making.

$$\boxed{\text{Raw Data} \longrightarrow \text{Statistical Analysis} \longrightarrow \text{Information} \longrightarrow \text{Decision}}$$

1.5.2 Artificial Intelligence and Machine Learning

Statistics provides an essential mathematical foundation for Artificial Intelligence (AI) and Machine Learning (ML). Many machine learning algorithms rely on concepts from probability and statistics. Statistical methods are used for:

- estimating model parameters;
- measuring uncertainty;
- identifying patterns in data;
- evaluating model performance;
- prediction and classification; and
- selecting appropriate models.

For example, classification algorithms estimate the likelihood that an observation belongs to a particular class.

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1.5.3 Probability and Uncertainty in Computing

Computer systems frequently operate under uncertain conditions. Probability provides a mathematical framework for representing and analyzing such uncertainty. For example, a computer network may experience packet loss, a server may fail unexpectedly, or a machine learning model may produce an uncertain prediction. Therefore,

$$\boxed{\text{Probability} \longrightarrow \text{Modeling Uncertainty}}$$

is an essential concept in many areas of computing.

1.5.4 Computer Networks

Statistical methods are widely used to analyze the performance and reliability of computer networks. Important quantities such as network traffic, packet arrival rates, transmission delays, and packet loss can be modeled using probability and statistical techniques. For example, if the response times of a server are recorded, statistics can be used to determine the average response time and the variability of the response time.

1.5.5 Cybersecurity

Statistics plays an important role in detecting abnormal activities and potential security threats. Statistical methods can be used to identify unusual patterns in network traffic, login behavior, system activity, and user behavior. For example, if a user's normal login behavior is statistically different from a particular login attempt, the activity may be classified as anomalous and investigated further.

1.5.6 Software Engineering

Statistical techniques are useful in software testing, quality assurance, reliability analysis, and defect prediction. For example, software developers can analyze historical testing data to estimate:

- software failure rates;
- defect frequencies;
- testing effectiveness;
- system reliability; and
- expected software performance.

Thus, statistics supports evidence-based software quality improvement.

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1.5.7 Computer Performance Analysis

Statistics is essential for measuring and evaluating the performance of computer systems. Performance indicators may include:

- CPU utilization;
- memory usage;
- execution time;
- response time;
- throughput; and
- energy consumption.

Statistical analysis helps engineers compare different systems, algorithms, and configurations under varying conditions.

1.5.8 Computer Vision and Image Processing

Computer vision systems process large amounts of visual data. Probability and statistical methods are used for image classification, pattern recognition, object detection, feature extraction, and uncertainty estimation. For example, an image classification system may assign probabilities to different possible classes before selecting the most likely class.

1.5.9 Natural Language Processing

Statistical methods have played a fundamental role in Natural Language Processing (NLP). Probability models can be used to analyze the frequency and relationships of words and linguistic patterns. Statistical techniques are applied in:

- language modeling;
- text classification;
- sentiment analysis;
- speech recognition; and
- machine translation.

1.5.10 Database and Information Systems

Statistics is useful for analyzing information stored in databases. Statistical summaries and data analysis can help identify trends, patterns, relationships, and anomalies within large datasets. Statistical techniques are also important in query optimization, data mining, and decision-support systems.

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1.5.11 Experimental Design and Algorithm Evaluation

Computer scientists frequently need to compare algorithms or system configurations. Statistical methods provide a scientific framework for conducting experiments and determining whether observed differences are meaningful. For example, suppose two algorithms have different average execution times. Statistical analysis can be used to determine whether the observed difference is statistically significant or simply the result of random variation.

1.6. Functions of Statistics

The following are the important functions of statistics:

a. It presents facts in a definite and systematic form.

b. It simplifies complex data into a meaningful form.

c. It facilitates comparison between different sets of data.

d. It helps in analyzing and interpreting numerical data.

e. It assists in forecasting future trends and outcomes.

f. It provides a basis for decision-making and policy formulation.

1.7. Limitations of Statistics

Although statistics is a powerful tool for collecting, analyzing, and interpreting data, it has several important limitations:

1. **Depends on Data Quality:** Incorrect, incomplete, or biased data may lead to misleading statistical conclusions.
2. **Deals Mainly with Quantitative Information:** Statistics primarily deals with numerical data. Qualitative aspects such as emotions, attitudes, and opinions may be difficult to measure accurately.
3. **Shows Association, Not Necessarily Causation:** A statistical relationship between two variables does not necessarily imply that one variable causes the other.
4. **May Hide Individual Differences:** Statistical measures such as averages and percentages describe groups but may not adequately represent individual cases.
5. **Involves Uncertainty:** Statistical results are often subject to sampling errors and other sources of variation; therefore, conclusions may not be completely certain.
6. **Can Be Misused or Manipulated:** Improper selection of data, statistical methods, samples, or graphical representations can produce misleading results.
7. **Requires Proper Interpretation:** Statistical results must be interpreted carefully in their appropriate context. Numbers alone may not explain the underlying reasons or mechanisms.
8. **Cannot Replace Human Judgment:** Statistics supports decision-making, but practical knowledge, experience, and logical reasoning are also necessary for sound decisions.

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Chapter-02
Collection of Data

In this chapter, we shall define some important basic concepts, which are needed to study and understand the subject statistics. Furthermore, we will discuss about different types of data collection tools and level of measurement.

Population

In statistics, a **population** refers to the complete set of individuals, objects, observations, or measurements that possess certain characteristics of interest and about which a statistical study is intended to draw conclusions.

> **Population is the totality or collection of all objects or individuals on which observations are taken on the basis of some characteritics in any field of investigation.**

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For examole,

1. If the objective is to study the academic performance of all undergraduate students of BAIUST, then the population consists of **all undergraduate students of BAIUST**.
2. If a researcher wants to investigate the lifetime of a particular type of electronic component, then the population consists of **the lifetimes of all components of that type produced under the specified conditions**.
3. If a study aims to determine the average height of students in a particular university, then the population consists of **the heights of all students in that university**.
4. If a company wants to examine the quality of its manufactured products, then the population consists of **all products manufactured by the company during the specified period**.

Kinds of Population

Statistical populations can be classified into the following types:

1. **Finite Population:** A population containing a fixed and countable number of units is called a finite population.

   Examples: The students enrolled in a university, the number of employees in a company, or the computers in a laboratory.

2. **Infinite Population:** A population containing an unlimited or infinitely large number of units is called an infinite population.

   Examples: The outcomes obtained from repeated tossing of a coin, or measurements generated by a continuous process over an unlimited period.

3. **Real Population:** A population consisting of physically existing and identifiable units is called a real population.

   Example: All students of a particular university.

4. **Hypothetical Population:** A population consisting of theoretically possible observations or outcomes is called a hypothetical population.

   Example: All possible outcomes obtained from repeatedly tossing a particular coin.

5. **Homogeneous Population:** A population whose units are similar with respect to the characteristics under study is called a homogeneous population.

   Example: A population consisting of students from the same academic year and program.

6. **Heterogeneous Population:** A population whose units differ significantly with respect to the characteristics under study is called a heterogeneous population.

   Example: The population of all students in a university when students from different departments and academic years are considered.

$$\text{Population} \longrightarrow \begin{cases} \text{Finite} & \text{Infinite} \\ \text{Real} & \text{Hypothetical} \\ \text{Homogeneous} & \text{Heterogeneous} \end{cases}$$

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2.1. Variable

In statistics, a **variable** is a characteristic, attribute, or measurable property of an individual or observational unit that can take different values from one unit, observation, or occasion to another.

In other words, a variable represents a characteristic whose value is not necessarily constant throughout the population under study.

Examples

1. **Age:** The age of individuals in a population may vary from one person to another. Hence, age is a variable.

2. **Height:** The heights of students in a class are not generally the same. Therefore, height is a variable.

3. **Marks:** The marks obtained by students in an examination constitute a variable because different students may obtain different marks.

4. **Monthly Income:** The monthly incomes of individuals may differ from person to person. Thus, monthly income is a variable.

5. **Number of Children:** The number of children in different families may take different values and is therefore a variable.

Variables can be classified into the following major types:

2.1.1 Qualitative (Categorical) Variable

A **qualitative variable**, also known as a **categorical variable**, describes non-numerical characteristics or attributes of individuals or objects.

Examples:

- Gender: Male, Female
- Blood group: A, B, AB, O
- Marital status: Married, Unmarried
- Department: Mathematics, Physics, CSE

2.1.2 Quantitative (Numerical) Variable

A **quantitative variable** is a variable whose values are expressed numerically. It represents characteristics that can be counted or measured. Quantitative variables are mainly classified into two types:

2.1.2.1 Discrete Variable

A **discrete variable** is a quantitative variable that can take only distinct and countable values, usually whole numbers.

Examples:

- Number of students in a class: $40, 41, 42, \ldots$
- Number of computers in a laboratory
- Number of defective products
- Number of children in a family

2.1.2.2 Continuous Variable

A **continuous variable** is a quantitative variable that can take any value within a given interval. Such variables are generally obtained through measurement. **Examples:**

- Height of a student: 165.5 cm
- Weight of a person: 62.75 kg
- Temperature: 36.8°C
- Time required to complete a task: 12.43 minutes

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2.1.3 Independent Variable

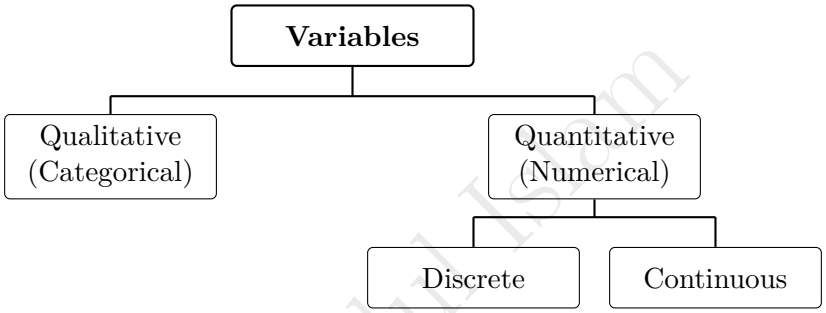
An **independent variable** is a variable whose value is controlled, manipulated, or selected by the researcher to investigate its effect on another variable.

Example: In a study of the effect of study time on examination performance, the **study time** is the independent variable.

2.1.4 Dependent Variable

A **dependent variable** is a variable whose value changes in response to changes in the independent variable. It represents the outcome or response being studied.

Example: In the study of the effect of study time on examination performance, the **examination performance** is the dependent variable.



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2.2. Scales of Measurement

World civilization is enriched by the idea of number and measurement. It was first felt in physical sciences but now a days it is spread nearly all branches of knowledge. At the end of the eighteenth century Lord Kelvin said, "when you can measure what you are speaking about and express it in numbers you know something about it; but when you can not express it in numbers, your knowledge is of a meager and unsatisfactory kind".

It is the belief of some researchers that if it is researchable then it must be measurable. If the measurement procedure in a statistical investigation is poor, the usefulness of the findings of the investigation will be severely affected. For every researcher, it becomes necessary to explain the variables under study as well as the level of measurements of the selected variables during the planning phase of the study. In the field of measurement it is also essential to know what the numbers actually imply. Now we want to explain the meaning of measurement.

Definition. Measurement. It is a process of assigning numbers to some characteristics or variables or events according to scientific rules.

In statistics, measurement has a special meaning. A measurement is obtained when a variable is

measured on experimental unit. All variables under statistical study can be measured. Since variables in study may be of different nature, they may have different level or scale of measurement. According to Stevens (1968), there are four levels or scales of measurements. They are

i) Nominal scale,

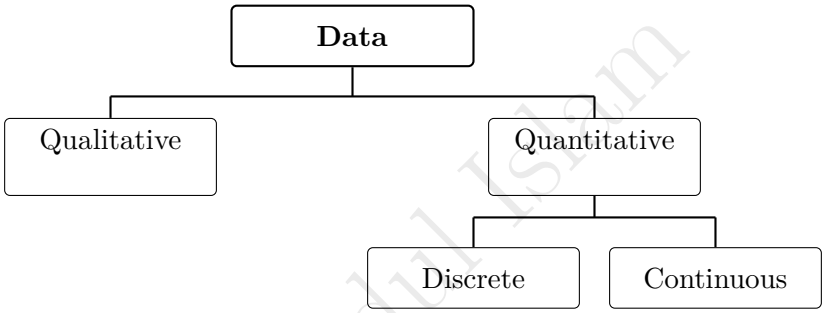
ii) Ordinal scale,

iii) Interval scale and

iv) Ratio scale.

Each scale is a way of assigning numbers to variables, and each has its limitation and degree of precision. According to different scale of measurement, variables are also called nominal, ordinal, interval and ratio.

The classification of the variables and measurement of scales is shown in following diagram.



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Qualitative variables are measured by nominal and ordinal scales where as quantitative variables are measured by interval and ratio scales. Now we shall define all the scales of measurement in turn with examples.

Definition. Nominal Scale. The scale of measurement by which we can classify and identify a qualitative variable according to different categories is called nominal scale.

The main purpose of this scale of measurement is to identify and then classify qualitative variables according to different categories. The variables measured by nominal scale are also called nominal variables and the data obtained by this scale of measurement are called nominal data. For example, the defective status (defective or not defective) of each of 100 piece of cloths produced by a factory.

Here are some examples of nominal scale:

1. **Gender** of a garment worker (Male, female),

2. **Colour** of eyes of a worker (Black, green, brown),

3. **Religion** of a worker (Muslim, Hindu, Buddhist, Christian),

4. **Marital** status of a worker (Single, married, widowed, divorced or separate).

The variables of the above example are bolded and are measured by nominal scale.

In example 1, a student is measured according to gender. He may be male or female. That is worker is first identified and the classified according to two categories. Similarly, a worker can be identified and classified according to the colour of eyes or religion or marital status.

It is the weakest level of measurement. These measurements are not measured natural numerical scale. But sometimes, they are denoted by number or alphabet or both. But this number cannot be used for mathematical operation such as addition, subtraction, multiplication or division. For example, the model or registration number of a car, the holding number of a house, class identity number of a student etc.

Definition. Ordinal Scale. The scale of measurement by which we can classify, identify and rank a qualitative variable according to different categories is called ordinal scale.

Ordinal scale is the second level of measurement. It possesses all the characteristics of the nominal scale. Some examples are given below:

1. **Grading** of a student (A, B, C, D).
2. **Size** of a worker (Tall, medium, short).
3. **Size** of a factory (Big, medium and small).
4. **Rating** of an Executive (Excellent, good, fair, poor).
5. **Economic** status of a citizen (Higher class, middle class, poor).
6. **Health** status of a worker (Excellent, good, poor).

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In example 1, students are classified into four categories according grade A, B, C, D. Here a student having letter grade A is better than a student having letter grade B. That is letter grade A, B, C, D are orderly classified. In example 2, the size of workers are measured as tall, medium or short. Here the values of the six qualitative variables are meaningfully ranked or ordered and then classified according to different categories. Here mathematical notation $>$ (greater than) or $<$ (less than) may be useful. But the equal, addition, subtraction, multiplication or division is not possible in ordinal scale. The data measured by this scale are also known as ordinal variables and the data obtained by the ordinal scale is an ordinal data. For example, efficiency (Excellent, good, fair, poor) of each 75 workers of a factory is an example of ordinal scale.

Definition. Interval Scale. The scale of measurement by which we can measure a quantitative variable numerically on experimental unit with zero as origin is called interval scale.

This is the third highest level of measurement. It preserves all the characteristics of nominal and ordinal scales. Some example of interval scale are:

1. Body temperature of a patient.

2. Marks obtained by students in an examination.

3. Calendar time.

The variables considered in the above three examples are quantitative. They are measured in numerical scale. Body temperature is measured in Fahrenheit or Centigrade. These two measurements have no absolute zero as origin. So zero body temperature does not mean "no temperature", 600 F is not twice the 300 F. But the difference between 600 F and 300 F is the same as 300 F and 300 F. Similarly, the marks obtained by a student have no absolute zero or origin. Mathematical operation such as addition and subtraction are only possible by this scale of measurement. The variables which are measured by this scale, are known as interval variables and the data obtained by this scale are called interval data. For example, the scores of a sample of 50 students in a class test.

Definition. Ratio Scale. The level of measurement is called ratio scale when a quantitative variable is measured numerically on experimental unit with absolute zero as origin.

Ratio scale is the highest level of measurement. All kinds of mathematical operations are possible in this scale. It preserves all characteristics of the previous three scales of measurements. Some examples of this scale of measurements are given below:

1. Age of a worker.

2. Weight of a worker.

3. Height of a worker.

4. Number of printing mistake per page of a book.

5. Number of children per family.

6. Number of defects of a product.

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All the variables in the above examples are quantitative. They are measured in natural numerical scale. Age, weight, height, number of printing mistakes, number of children and number of defects are all measured not only in natural numerical scale, they have also absolute zero as origin. Zero age of a worker means no age, zero weight or zero height means no weight or no height.

The variables measured by this scale of measurement are called ratio variable and the data obtained by this scale are called ratio data. For example, the age for each of 50 workers of a factory.

Example Identify the scale of measurement on which the following underlined variables are measured:

1. **Colour** of a car entering in the parking lot.

2. **Model** of a car made by the fort motor company.

3. **Holding number** of a house in Dhaka city.
4. **Smoking habit** of a student.
5. **Types of houses** for the University teachers (A, B, C, D).
6. **Socio-economic status** of a citizen (upper, upper-middle, lower-middle, lower).
7. **Size** of an apple in a basket (big, medium, and small).
8. **Quality** of each student (Best, average, poor) in a class of 100 students.
9. Body **temperature** of a patient.
10. The **grade point average** (GPA) obtained by each student of a class of 100 students.
11. Room **temperature** of a house in winter.
12. **Number of children per family** of 100 families selected from an area.
13. **Printing mistakes per page** of a book.
14. **Height** of a cricket player.
15. **Weight** of a soldier in the Bengal Regiment.
16. **Political affiliation** of a citizen.

Solution. 1. Nominal Scale, 2. Nominal scale, 3. Nominal scale, 4. Nominal scale, 5. Ordinal Scale, 6. Ordinal Scale, 7. Ordinal Scale, 8. Ordinal Scale, 9. Interval scale, 10. Interval scale, 11. Interval scale, 12. Ratio scale, 13. Ratio scale, 14. Ratio scale, 15. Ratio scale, 16. Nominal scale.

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2.3. Meaning of Data

The term *data* refers to a collection of facts, figures, observations, measurements, or information gathered for a specific purpose. Data may be obtained through observation, experimentation, surveys, interviews, or other systematic methods. It serves as the basic input for statistical analysis and decision-making. In simple terms, **data are raw facts and figures collected for analysis and interpretation.**

Usually data are the numerical results of scientific measurements. For example, it could be the

i. Income of workers,
ii. IQ scores of students,
iii. Examinations marks of students in a class test, etc.

2.4. Types of Data

Statistics is the science of data and that data are obtained by measuring the values of one or more variables on the experimental unit in the population or sample. Broadly speaking data may come from many ways such as

1. According to origin,
2. According to variable,
3. According to scale of measurement,
4. According to time,
5. According to sources.
6. According subjects

2.4.1 According to origin

Data may be obtained from

i) Population and

ii) Sample.

Definition. Population Data. When data originate from each experimental unit of the population on the basis of some variable is called population data. It is also called census data.

Following are the examples of population data:

1. Heights of 500 students of a department may be population data if these 500 students are the experimental units of the population.
2. Salary of all the workers of a factory,
3. Marks obtained by all students of a class in a test, etc.

Definition. Sample Data. Data are called sample data when they are obtained from each experimental unit of a sample.

The following are examples of sample data:

1. Salary of 50 workers out of 500 is a sample data,
2. Ages of 40 workers randomly selected from 500 workers of a garment.
3. The health condition of 50 workers of a factory randomly selected from 1000 workers.

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2.4.2 According to variable

Again all data can be classified as one of two general types according to variable as

1. Qualitative data and
2. Quantitative data.

Definition. Qualitative data. Qualitative data are obtained when a qualitative characteristic is measured on each experimental unit.

Qualitative data cannot be measured on a natural numerical scale. They can only be classified into one of a group of categories.

Examples of qualitative data include:

1. The defective status (defective or non-defective) of each of 100 bulbs produced by a Philip's company,
2. Working Efficiency of 60 workers of a factory (Excellent, good, satisfactory, poor),
3. Gender (male, female) of 5000 workers of a garment factory,
4. The eyesight (excellent, good, poor) of a sample of 100 male students of Chittagong University.

Sometimes qualitative data are called categorical data or count data. Often, we assign arbitrary numerical values to qualitative data for the case of computer entry and analysis. But these assigned numerical values are simply codes. They cannot be meaningfully added, subtracted, multiplied or divided. For example, we might code Defective = 1, Non-defective = 2. Similarly, efficiency of a worker might rank from excellent = 1 to poor = 4. These are simply arbitrary selected codes for the categories and have no use beyond that.

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Definition. Quantitative data. Quantitative data are obtained when a quantitative variable is measured on each experimental unit. Quantitative data are measured on a naturally occurring numerical scale.

The following are the examples of quantitative data:

1. The daily production of a factory for last one month.
2. The monthly export of tea for the first six months of the financial year 2007-2008.
3. The body temperature of 60 patients of clinic.
4. The number of family members of 150 workers of a factory.
5. The GPA scores of 200 S.S.C. students of Chittagong University School for the year 2008.

Again there are two types of quantitative data. They are

1. Discrete data and
2. Continuous data.

Definition. Discrete data. Data generated from a discrete variable is called discrete data.

Examples.

(1) Number of babies born per day for last six months of the year 2007 in Chittagong Medical College,

(2) Numbers of workers absent per day for last two months of a garment industry,

(3) Number of children per family of 150 families of a village.

Definition. Continuous data. Data originated from a continuous variable is called continuous data.

Examples.

1. Weight in pound of 50 students of a class,
2. Body temperature of 70 patients of a hospital,
3. Heights in feet of 100 first year students of Chittagong University,
4. Weight in kg of 100 finished product of a factory.

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2.4.3 Data according to scale of measurement

According to different level of measurements data are also known as

i) Nominal data,

ii) Ordinal data,

iii) Interval data, and

iv) Ratio data.

(i) Nominal data. A set of data is called nominal data when it is originated from a variable which is measured on nominal scale. Some examples of nominal data:

1. The defective status (defective or not defective) of each of 100 garments items manufactured by a factory,
2. Colour of each of 100 cars entering in a parking lot,

3. Types of ice creams (cup, choc-bar, lemon, and orange) available in a cooling corner.

(ii) Ordinal data. A set of data is called ordinal data when it is originated from a variable which is measured on ordinal scale. The following are some of its examples:

1. The size of a car (subcompact, compact, mid-size or full-size) rented by each of a sample of 40 business travelers,
2. The letter grade (A, B, C, D) obtained by 100 student of a school in H.S.C. examination for the year 2008,
3. The size (large, medium and small) of each apple of 150 apples in a basket.

(iii) Interval data. A set of data is called interval data when the variable is measured on interval scale. The following are the examples of interval data:

1. The grade point average (GPA) obtained by a sample of 50 students got admission in BBA course for the session 2007-2008,
2. The temperature (in degree Celsius) at which each in a sample of 40 pieces of heat-resistant plastic begins to melt.

(iv) Ratio data. A set of data is called ratio data when a variable is measured on ratio scale. Some examples of ratio scale are:

1. The number of printing mistakes per page of a book of 800 pages,
2. The weight of each of 150 cartons ready to be shipped,
3. The height of 50 students in a class.

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2.4.4 Data according to time

According to time data may be classified as:

i) Time series data;

ii) Cross-section data; and

iii) Panel data.

(i) Time series data. A time series data is a set of ordered observations on a quantitative characteristic of an individual or collective phenomenon taken at different points of time. Although it is not essential, it is common for these points to be equidistant in time.

1. Year-wise production of a firm for last 15 years,
2. Month-wise price of potatoes for last six years,
3. Population of a country for last fifteen years,

4. Productions of a factory for last ten years etc are some examples of time series data.

(ii) Cross-section data. Data are called cross-section when they are collected from the experimental units at a particular period of time.

1. Salary of the workers of a factory for the month of July, 2005.
2. Height of the students of a class,
3. Weight of patients of a clinic, etc. are the examples of cross-section data.

(iii) Panel data. This is the mixture of time series and cross-section data.

1. Year-wise prices of different food stuffs for last ten years,
2. Income and expenditure of 50 workers of a factory for last 10 years,
3. Month-wise productions of different items of a firm for last 15 years are some examples of panel data.

2.4.5 According to sources

According to source data are known as:

i) Primary data; and

ii) Secondary data.

2.4.6 According to subject

Data may be business data, agriculture data, medical data, meteorological data, economic data etc.

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2.5. Sources of Statistical Data

Data constitute the foundation of statistical analysis and interpretation. They are the raw materials of any kind of statistical analysis. The success of any statistical investigation depends on the quality of data. Hence the first step in statistical work is to collect data through proper data collection technique. Data are collected from two important sources, namely

i) Primary source and

ii) Secondary source.

Depending on the source, data are called primary data and secondary data.

2.5.1 Primary data

Definition Primary data are those data which are collected by the investigator himself or by any research institution for the purpose of some specific inquiry or study.

Such data are collected afresh and for the first time and thus happen to be original in character and are generated by surveys conducted by individuals or research institutions. For example, if a researcher is interested to know the salary structure of the female workers of a garment factory, he must take a survey and collect data on salary from the female workers of the relevant garment factory. The data so collected would be considered as a primary data.

Data collected by different government, public, private organizations, research bodies, research scholars, NGO's for their official records and research purpose from the field directly are primary data. Bangladesh Bureau of Statistics (BBS) is the main primary sources of government data.

2.5.2 Secondary data

Definition When an investigator uses data which have already been collected by others, such data are called secondary data.

Such data are primary data for the agency that collect them, and become secondary for some one else who uses these data for his own purposes. The secondary data can be obtained from Journals, reports, government publications, publications of professional and research organizations etc. For example, if a businessman wants to study the movements of the share indices of different companies, he can get the required information or data from the share market related publications as secondary data. This means the primary data collected by an agency or organization; constitute the secondary data in the hands of other agencies.

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All the data collecting methods are broadly divided into two categories viz.

i) Methods of collecting Primary data, and

ii) Methods of collecting Secondary data.

2.6. Methods of Collecting Primary Data

The collection of data may range from a simple observation to a large-scale survey on any defined population. The tools and techniques to be employed to collect data depend largely on the objectives of the study, the research design and the availability of the time, money and trained personnel. There are several primary data collecting methods each with its own advantages and disadvantages. The important primary data collecting methods are:

1. Through Interview,

2. Through Questionnaires,

3. Through Schedules,
4. Through Local agents,
5. Through observations,
6. Through experimentation.

A brief description on each of the above mentioned methods is provided below.

2.6.1 Through Interview

It is one of the important methods of collecting primary data. Interview methods can be categorized as:

1. Unstructured Interview,
2. Structured Interview,
3. Direct Personal or Face to face Interview,
4. Indirect Personal Interview,
5. Telephone Interview,
6. Computer-Assisted Interviewing.

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(i) Unstructured Interview

By this method, the interviewer is free to ask any question to the respondents relating to the objective of the study. The main purpose of this method is to bring some preliminary issues to the surface so that the researcher can determine what variables need further in-depth investigation. Usually open ended questions are asked to the respondents. For example: Tell me something about your unit and department, and perhaps even the organization as a whole, in terms of work, employees, and whatever else you think is important.

(ii) Structured Interview

By this method, the interviewer has a list of predetermined questions to be asked to the respondents either personally, through the telephone, or through the medium of a PC. Here the same questions will be asked to everybody in the same manner. This is particularly important when a team of trained interviews conducts the survey.

(iii) Direct personal or face to face Interview

Under this method, the investigator presents himself personally before the respondents and ask questions in a face-to-face contact and obtains a first hand information. This sort of interview is useful in the investigation of a specific issue or opinion survey.

Merits

1. The data collected by this method is very reliable and accurate,
2. The response rate is almost 100%.

Demerits

1. The method is not suitable for a large scale survey.
2. It is costly and time consuming.

Recommendation. The method is suitable when the field of enquiry is small and a greater degree of accuracy is required.

(iv) Indirect Personal Interview

Under this method, instead of directly approaching the respondents, the investigator interviews several third persons who are directly or indirectly connected with the subject matter of the enquiry and who are in possession of the requisite information. For example, data regarding drug addicted, gamblers, smugglers, fighting between two groups of students etc. are successfully obtained by this method. This method is followed by the enquiry committees and commissions appointed by the government, universities and different organizations.

Merits

1. The method is less costly and less time consuming then the direct personal investigation.
2. Under this method, the enquiry can be formulated and conducted more effectively and efficiently

Limitation. The success of this method depends upon

1. The representative character of the witness.
2. The personal knowledge of the witness about the subject-matter of enquiry.
3. The ability of the investigator

The method is very popular in every day life.

(v) Telephone Interview

Under this method, the investigator contacts respondents on telephone and collects information from them. The method is best suited when information from a large number of respondents spread over a wide geographical area is to be obtained quickly, and the probable duration of each interview is, say, 10 minutes or less. Many market surveys are conducted through structured telephone interviews, particularly in developed region. The method is more convenient than personal interview.

Merits

Shahidul Islam

1. It is less costly and less time-consuming and can be applied even to extensive fields of enquires.

2. The method is more convenient than personal interview.

Limitation

1. The method excludes those who do not have telephone as also who have unlisted telephones.

2. The main limitation of this method is that the number of questions should be limited.

Most opinion surveys are done by this method.

(vi) Computer-Assisted Interviewing (CAI)

Under this method, questions are flashed onto the computer screen and interviewer can enter the answers of the respondents directly into the computer. There are two types of computer-assisted interview programs: Computer-assisted telephone interviewing (CATI) and computer-assisted personal interviewing (CAPI).

- **Computer-Assisted Telephone Interviewing.** Through this method the interviewer can collect data from people all over the world since PC is networked into the telephone system. The PC monitor promotes the questions with the help of software and the respondent provides the answers. The computer selects the telephone number, dials and places the responses in a file. The data are analyzed later.

- **Computer-Assisted personal Interviewing.** This method involves big investments in hardware and software. CAPI has an advantage in that it can be self-administered, that respondents can use their own computers to run the program by themselves once they receive the software and enter their responses, thereby reducing errors in recording.

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2.6.2 Collection of data through questionnaires

This method of data collection is quite popular, particularly in case of big enquiries. Under this method, the investigator prepares a questionnaire containing a number of questions relating to the objectives of the study. These questionnaires are sent by post to the respondents together with a polite covering letter explaining in details, the aims and objectives of collecting the information, and requesting the respondents to cooperate by furnishing the correct replies and returning the questionnaire duly filled in. In order to ensure quick response, the return postage expenses and some monetary incentives are given to the respondents by investigator. Now a day, questionnaires are also sent to the respondents through e-mail addresses. This method is usually adopted by the research workers, private individuals, private and public organizations and even by governments.

Merits

1. By this method, a large field of investigation may be covered at a very low cost.

2. This is the most economical method in terms of time, money and manpower.

3. Personal bias of the investigators or enumerators are eliminated by this method.

Limitations

1. The method can be applied to the educated respondents only.

2. The return rate of the filled questionnaires are not always satisfactory.

The method is suitable when the area of the investigation is large and the respondents are educated.

2.6.3 Collection of data through Schedule

Under this method, instead of sending the questionnaire through post, the investigator appoints local agents known as enumerators, who go to the respondents personally with the questionnaire called schedules, ask them the questions given there in, and record their replies. The enumerator should be given proper training to perform their job well. This method of data collection is very useful in extensive enquires and can lead to fairly reliable results.

Merits

1. The information collected through this method is more reliable.

2. Non-response rate is very low compared to questionnaire method.

3. The method can be applied to illiterate respondents.

Limitations

1. The method is more expensive and can be used by financially strong bodies or institutes only.

2. It is more time-consuming then mailed questionnaire method.

3. Skilled and well-trained enumerators needed to get reliable data.

Population census all over the world is conducted through this method.

2.6.3.1 Requirements of a good questionnaire

Questionnaire is a set of questions relating to the objective of the study. The success of questionnaire method of collecting information depends largely on the proper designing of the questionnaire. Designing of questionnaire, requires a high degree of skill and experience on the part of the investigators. There is no hard and fast rules for designing or framing a questionnaire. While developing a questionnaire, the researcher has to be very clear on the following issues:

1. The number of questions should be related to the objective of the study.

Shahidul Islam

2. The size of the questionnaire should not be too large.
3. Questions should be logically arranged.
4. Questions should be clear, brief, unambiguous, non-offending, and courteous in tone, corroborative in nature and to the point.
5. Questions of sensitive nature should be avoided.
6. The questions should be simple and easy to understand.
7. The questions should be so designed that the respondents can easily comprehend and answer them.
8. Questions involving mathematical calculation should not be asked.
9. If a particular questions need clarification, it should be explained by way of a footnote.
10. The questionnaire should provide necessary instructions to the respondents.
11. The answers of most questions should be of multiple choices. That is closed questions should be more then the open questions.

2.6.3.2 Difference between questionnaire and schedule methods

Both questionnaire and Schedule are popularly used methods of collecting data in research survey. There is much resemblance in the nature of these two methods and this fact has made many people to remark that from practical point of view, the two methods can be taken to be the same. But from the technical point of view there is difference between the two. The important points of difference are:

1. The questionnaire is sent to the respondents in a covering letter whereas the schedule is filled out by the research workers or by the enumerators.
2. Mailed questionnaire method is cheaper and economical than the schedule method.
3. Non-response rate is higher in case of questionnaire method than schedule method.
4. Questionnaire method is slow compare to the schedule method.
5. Questionnaire can be used in care of literate respondents, whereas schedule method can be even for illiterate respondents.
6. In case of questionnaire, it is not always clear who replies, but in case of schedule the identity of respondent is known.
7. The success of the questionnaire method depends on the quality of the questionnaire, but in case of schedules much depends upon the honesty and competence of enumerators.

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2.6.4 Collection of data through local agent

Under this method, the information is not collected formally by the investigators, but by local agents, commonly known as correspondents, is appointed in different parts of the area under investigation. These agents collect information in their areas and transmit the same to the investigator. They apply their own judgments as to the best method of obtaining information. This method is usually employed by newspaper or periodical agencies who require information in different fields like economic trends, business, stock and share markets, sports, politics and so on.

Merits

1. The method is very cheap and economical for extensive investigations.
2. The required information can be obtained expeditiously since only rough estimates are required.

Limitations

Information collected through this method may be subject to bias due to personal choice of local agents.

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2.6.5 Data collection through observation

Observation is one way to collect primary data. Observation is a purposeful, systematic and selective way of watching and listening to an interaction or phenomenon as it takes place. There are many situations in which observation is the most appropriate method of data collection; for example, when you want to study the animal behaviour, behaviour of a group of people etc. There are two types of observation:

(i) Participant observation, and

(ii) Non-participant observation.

(i) Participant observation

It is a method when the researcher participates in the activities of the group being observed in the same manner as its members, with or without their knowing that they are being observed. For example, one can study the economic, social, cultural behaviour of a class of people or race by this method.

(ii) Non-participant observation

When the researcher will not participate in the activities of the group but remains a passive observer, watching and listening to its activities and drawing conclusions from this. For example, a researcher wants to study the functions carried by a nurses in a hospital or a manager of a firm. As an observer, a researcher could watch, follow, and record the activities as they are performed. After making a number of observations, conclusions could be drawn about the functions of a nurse's carry out in the hospital or the manager in the firm.

In this method data are also collected by observation through mechanical devices such as the reading of X-ray films, E.C.G. etc. Meteorological data such as rainfall, temperature, humidity etc.; laboratory data of blood, urine, stool etc. are obtained by observations.

2.6.6 Collection of data through experimentation

In natural sciences like physics, chemistry and in biolog-al sciences like botany, zoology, biochemistry, microbiology and pharmacy, data are obtained when treatments are applied on experimental units in a controlled laboratory experiment. There are some important basic experimental designs such as (i) completely randomized design, (ii) randomized block design and (iii) Latin square design are used to obtain a set of good experimental data.

2.7. Sources of Collecting Secondary Data

There is no method of collecting secondary data like the method of collecting primary data. We can get secondary data from different sources. Before using secondary data the investigator should examine the following aspect:

1. Whether the data are suitable for the purpose of investigation.
2. Whether the data are adequate for the purpose of investigation.
3. Whether the data are reliable.

The chief sources of secondary data may be, broadly classified into the following two groups:

(i) Published sources

(ii) Unpublished sources

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(i) Published Sources

There are a number of national organizations and also international agencies which collect and publish statistical data relating to business, trade, labor, price, consumption, production etc. These publications of the various organizations are used for sources of secondary data. Some of these published sources are as follows:

- Government, semi-government organizations, municipalities, local bodies, universities publish a lot official statistics in report form. Bangladesh Bureau of Statistics(BBS) is the main sources of government official statistics.
- International organization like UNO, FAO, UNPD, ILO, IMF, UNESCO, UNEP, WHO, USAID, etc have got their official, publications containing valuable statistical information.
- Trade Journal, technical and financial journals as well as news papers contain a good deal of statistical data.

(ii) Unpublished sources

The records maintained by private firms or business houses who may not like to release their data to any outside agency, the researches carried out by the research scholars in the Universities or Research Institutes may also provide useful statistical data.

2.8. Selection of Appropriate Method

Selection of appropriate method for data collection. One must always remember that each method of data collections has its uses and none is superior in all situations. For instance, telephone interview method may be considered appropriate if funds are restricted, time is also restricted and the data is to be collected in respect of few items with or without a certain degree of precision. This method is appropriate for conducting a survey. In case funds permit and more information is desired, personal interview method may be said to be relatively better. In case of time is ample, funds are limited and much information is to be collected with no precision, then questionnaire method can be regarded more reasonable. When funds are ample, time is also ample and much information with no precision is to be collected, then either direct personal interview or the questionnaire or the joint use of these two methods may be considered as an appropriate method of collecting data. When a vast geographic area is to be covered, the use of questionnaire supplemented by personal interviews will yield more reliable results.

With greater technological advancement and a reduction of hardware and software costs, computer-assisted interviews promise to become a primary method of data collection in the future. The secondary data may be used in case the researcher finds them reliable, adequate and appropriate for his research. Thus, the most desirable approach with respect to the selection of the method depends on the nature of the particular problem and on the time and resources available along with the desired degree of accuracy.

2.9. Processing of Data

The data, after collection has to be processed through editing and coding.

2.9.1 Editing

Once the set of data have been collected, it is necessary to process them for proper presentation. Editing of data is required as preparatory work before the tabulation and statistical analysis is carried out. This is quite a difficult job and requires a great deal of skill and experience. While editing primary data the following consideration need attention:

1. The data should be complete.
2. The data should be consistent.
3. The data should be accurate.

4. The data should be homogenous.

It should be noted that these days computer is extensively used to edit data.

2.9.2 Coding

When the data is to be processed by computer, it must be coded and converted into the computer language. For some qualitative characteristics, the code numbers can be assigned and identified. For example, to a question, "Are you married or single?" a code of 1 can be assigned to the qualitative answer "married" and a code of 0 can be assigned to the answer "single". This coding job should be accomplished while editing the data.

The next step is to code the responses. Nowadays Scanner sheets are available to collect data through questionnaire. Such sheets facilitate the entry of the responses directly into the computer without manual keying in of the data. Raw data can also be entered through any software program say SPSS, Excel etc. Statistical analysis can also be done with the help of these software.

Sample Questionnaire

Questionnaire

A study entitled 'Determinants of Credit Cards Users in Cumilla City' was prepared by a graduate student for the partial fulfillment of his M.S. degree. The main objectives of the study are as follows:

- To know the information of the cardholder's about their social and educational status.
- To know the information of the cardholder's about their income and expenditure behaviour.
- To know the idea about the age structure of the credit card users.
- To know the information about the income group who uses credit cards
- To know the information about the types of cards they use.
- To know the idea about the satisfaction level of the cardholders.
- To know the problems they frequently faced.
- To know the information about the monthly average transition of money by the cardholders
- To take idea about the reason of using credit card.
- To acquire knowledge about credit card system and operational system of credit card.

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Questionnaire

1. Name:
2. Age:

Year	Month	Day

3. Sex: ☐ Male ☐ Female
4. Educational qualification:

   ☐ Graduate and above ☐ HSC

   ☐ SSC ☐ Others
5. Marital status: ☐ Single ☐ Married ☐ Divorce
6. Is Spouse Employed? ☐ Yes ☐ No
7. Where do you reside? (For example: Agrabad)
8. Your resident status:

   ☐ Own ☐ Rented ☐ Company provided

   ☐ Government provided
9. Your job status:

   ☐ Government employee ☐ Businessman

   ☐ Retired ☐ Student ☐ Others
10. Car ownership: ☐ Yes ☐ No
11. Family economic status: ☐ Lower ☐ Middle ☐ Higher
12. Monthly Income: _____
13. Monthly Expenditure: _____
14. Spouse Income (if any): _____
15. Total Income (Both): _____
16. Do you use debit card: ☐ Yes ☐ No
17. Which Bank's Credit card do you use? _____
18. Type of your card:

    ☐ Local Classic/Silver ☐ Local Gold ☐ Dual Classic/Silver

    ☐ Dual/International Gold ☐ Others (Please specify)

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19. Credit limit: _____

20. Average monthly transactions by credit card: _____

21. How many Credit card you use?

    ☐ One ☐ Two ☐ Three or more

22. You use your International credit card more in:

    ☐ Home country ☐ Abroad

23. For which purpose you use the credit card frequently?

    ☐ Business ☐ Purchases ☐ Smartness ☐ 24 hours service

24. You have taken card: ☐ Willingly ☐ By influence

25. Which problem you face usually?

    ☐ High interest rate ☐ Booth facilities ☐ Others (please specify)

26. What is your satisfaction status?

    ☐ Satisfied ☐ Highly satisfied ☐ Not satisfied

27. Do you want to continue in future? ☐ Yes ☐ No

28. Do you encourage your friends or relatives to take credit card?

    ☐ Yes ☐ No

Thank you for your co-operation

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Questions

1. Explain the main sources of data. Define primary and secondary data.
2. Explain the differences between primary and secondary data. State the important sources of secondary data.
3. Explain the questionnaire and schedule methods of collecting primary data. Distinguish between the questionnaire and schedule methods.
4. Explain the concept of secondary data. Describe the important sources of secondary data and state the precautions necessary for using secondary data.
5. Explain the meaning of editing data. Describe the procedure followed in editing primary data.
6. Describe the different types of interview methods used for collecting primary data.
7. Explain the factors that should be considered when selecting an appropriate method for collecting data.

8. Define population, sample, experimental unit and variables with suitable examples.

9. Identify the experimental units and variables of the following:

   (a) Gender of a garment worker,

   (b) Number of errors on a midterm exam,

   (c) Body temperature of a patient,

   (d) Age of a first year student,

   (e) Colour of a car entering the parking lot,

   (f) Number of typing errors per page of a manuscript,

   (g) Age of a cancer patient.

10. Distinguish between qualitative and quantitative variables. Identify each of the following variables as qualitative or quantitative:

    (a) The most frequent use of your microwave oven (reheating, defrosting, warming, other),

    (b) The number of consumers who refuse to answer a telephone survey,

    (c) Number of unregistered taxicabs in a city,

    (d) Number of defective items per box each of which contains 10 items,

    (e) Size of a factory (large, medium and small),

    (f) Rating of performance of a newly elected politician (excellent, good, fair, poor),

    (g) Number of deaths in every year in the Chittagong University campus,

    (h) The size of a firm (big, medium and small).

11. Define discrete and continuous variables with examples. Identify following variable as continuous or discrete:

    (a) The number of automobile accidents per year in Dhaka city,

    (b) The length of time in minutes to complete a phone call,

    (c) The number of telephone calls made from a telephone booth,

    (d) The quantity of paddy produced per acre,

    (e) The quantity of milk produced in a year by a herd of cows,

    (f) Ages of workers of a factory,

    (g) Number of claims received by an insurance company in every year,

    (h) Systolic blood pressure of a patient.

12. What do you mean by a variable? Discuss different types of variables with examples.

13. Explain with examples how does a population differ from a sample.

14. Illustrate the difference between discrete and continuous variables with examples.

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15. What do you mean by scale of measurement? Discuss different levels of measurements with suitable examples.

16. Identify the experimental units, variables, types of variables and their scales of measurements on which the following variables are measured:

    (a) Religion of an employee (Muslim, Hindu, Buddhist, Christian),

    (b) Official status of an employee (1st class, 2nd class, 3rd class, 4th class),

    (c) Monthly Salary of an employee,

    (d) Monthly expenditure of an employee,

    (e) Number of family members of an employee,

    (f) Humidity at Chittagong city,

    (g) Number of children per family of an employee,

    (h) Age of an employee,

    (i) Height of an employee,

    (j) Weight of an employee,

    (k) IQ score of candidates applied for a post,

    (l) Name of the students of CSE.

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